

# Predicting Income Level from U.S. Census Data

*A supervised learning study on the UCI Adult / Census Income dataset*

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Logistic Regression vs. Random Forest • Education, Age & Hours Worked as Key Predictors

# The Problem

## Research question

Which demographic and employment attributes are most strongly associated with earning over \$50,000/year — and how well can standard classifiers predict income class from them?

## Why it matters

The Adult dataset (Kohavi, 1996) is one of the most-used ML benchmarks, but later work shows it also encodes real historical bias — so strong accuracy alone isn't the full story.

**32,561**

census records  
(1994 U.S. Census)

**14**

demographic &  
employment attributes

**~25%**

of individuals  
earn >\$50K/year

# Method

1

## Clean

Drop rows with missing values ("?"). 32,561 → 30,162 records.

2

## Prepare features

Standardize 5 numeric features; one-hot encode 7 categorical features.

3

## Split

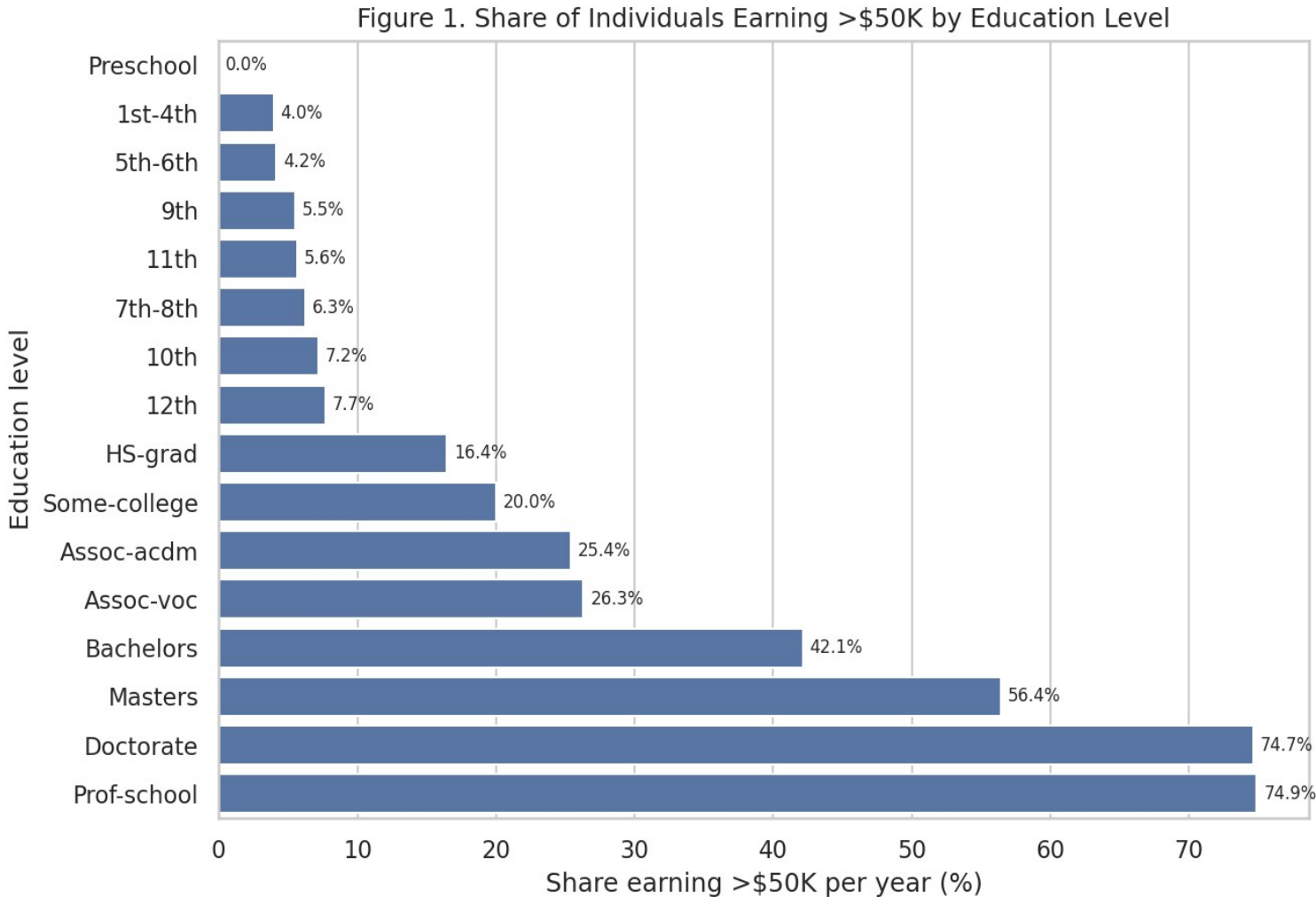
Stratified 80/20 train-test split on the income label.

4

## Model & evaluate

Train Logistic Regression and Random Forest; compare accuracy, precision, recall, F1, ROC AUC.

# Results: Education is the strongest visible driver



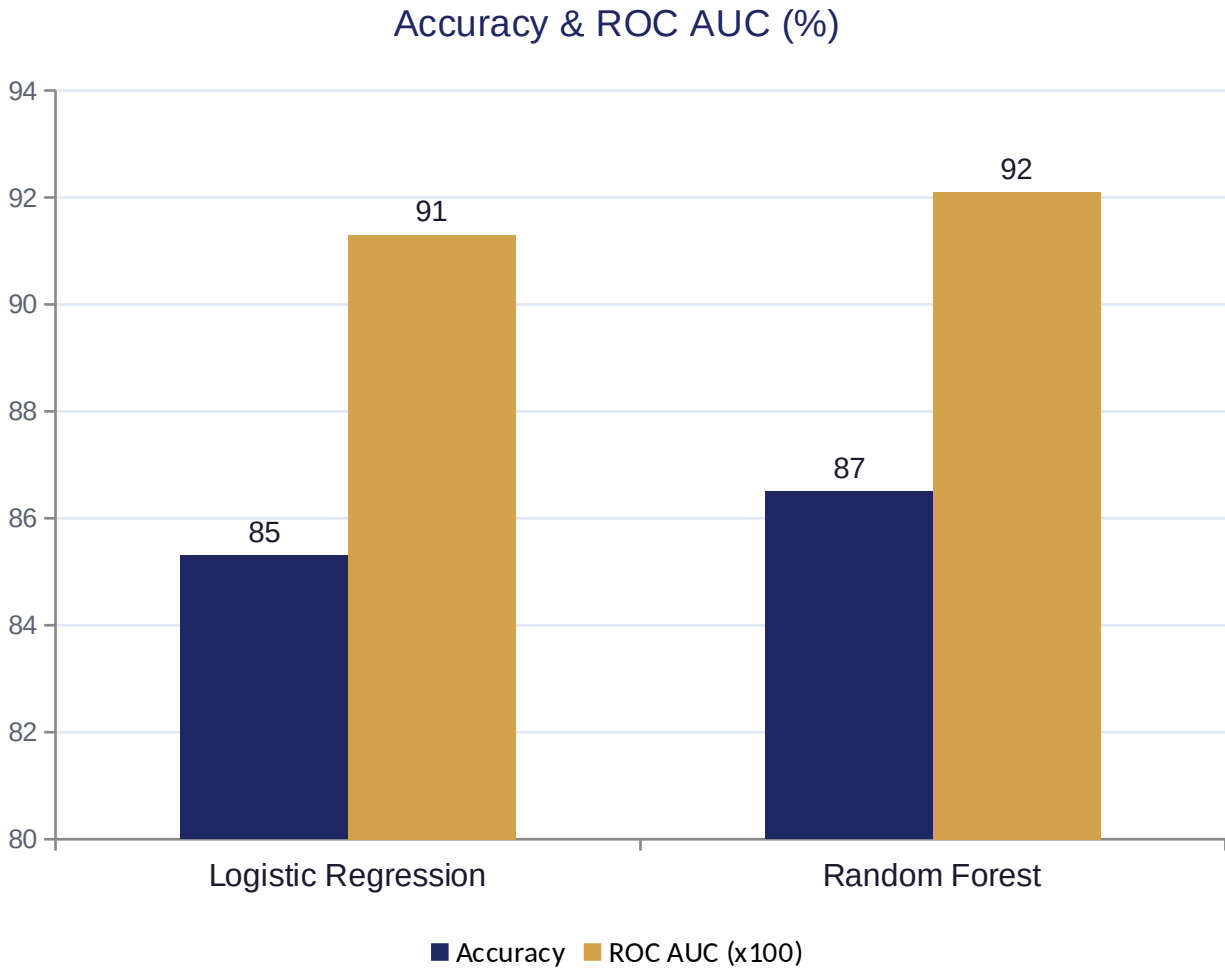
**Prof-school / Doctorate**

~75% earn >\$50K

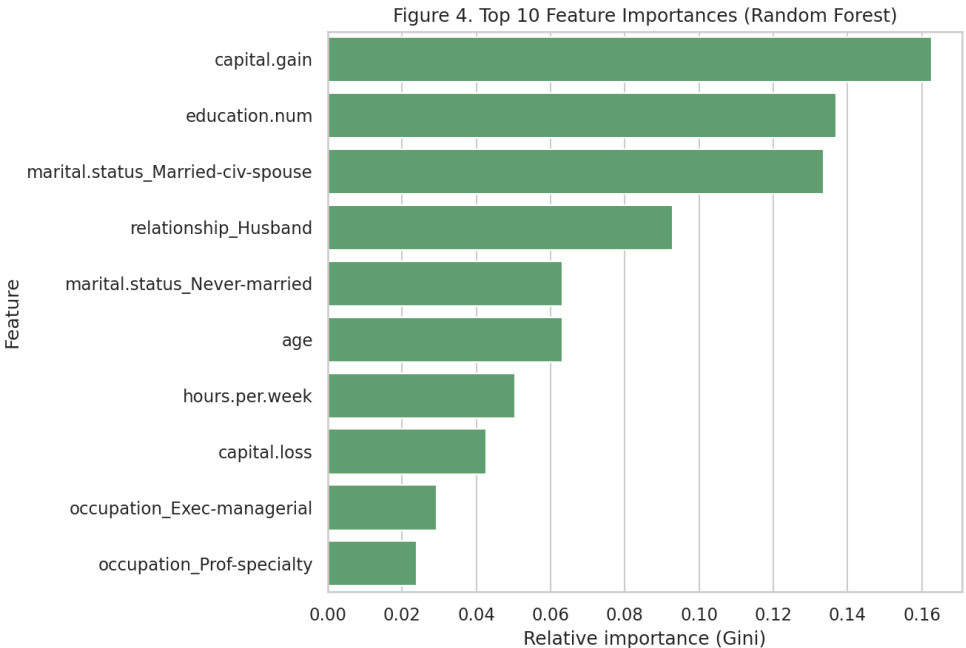
**No diploma**

<8% earn >\$50K

# Results: Random Forest edges out Logistic Regression



| Model               | Acc.  | Prec. | Recall | F1    | AUC   |
|---------------------|-------|-------|--------|-------|-------|
| Logistic Regression | 0.853 | 0.751 | 0.615  | 0.676 | 0.913 |
| Random Forest       | 0.865 | 0.821 | 0.585  | 0.683 | 0.921 |



# Takeaway

Education, age, hours worked, and relationship status are the dominant correlates of income in this dataset — and a Random Forest model captures them only slightly better than a simple Logistic Regression.

*But strong accuracy isn't the whole picture: prior fairness research on this same dataset shows these models can reproduce historical gender and demographic bias, so results should inform, not dictate, real decisions.*

**86.5%**

Best model accuracy  
(Random Forest)